



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Surds

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Your notes

Simplifying Surds

Surds & Exact Values

What is a surd?

- A surd is the square root of a non-square integer
- Using surds lets you leave answers in exact form
 - e.g. $5\sqrt{2}$ rather than 7.071067812...

THESE ARE SURDS:

$$\sqrt{2} \quad \sqrt{13} \quad \sqrt{99} \quad \sqrt{201}$$

THESE ARE NOT SURDS:

$$\sqrt{16} \quad \sqrt{\frac{2}{3}} \quad \sqrt{6.23}$$

16 IS A SQUARE NUMBER
 $\sqrt{16} = 4$

$\frac{2}{3}$ AND 6.23
AREN'T INTEGERS

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How do I do calculations with surds?

- **Multiplying surds**
 - You can multiply numbers under square roots together
 - $\sqrt{3} \times \sqrt{5} = \sqrt{3 \times 5} = \sqrt{15}$
- **Dividing surds**
 - You can divide numbers under square roots



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$$\frac{\sqrt{21}}{\sqrt{7}} = \sqrt{21} \div \sqrt{7} = \sqrt{21 \div 7} = \sqrt{3}$$

Factorising surds

- You can factorise numbers under square roots

$$\sqrt{35} = \sqrt{5 \times 7} = \sqrt{5} \times \sqrt{7}$$

Adding or subtracting surds

- You can only add or subtract multiples of "like" surds
 - This is similar to collecting like terms when simplifying algebra

$$3\sqrt{5} + 8\sqrt{5} = 11\sqrt{5}$$

$$7\sqrt{3} - 4\sqrt{3} = 3\sqrt{3}$$

- However $2\sqrt{3} + 4\sqrt{6}$ cannot be simplified
- You cannot add or subtract numbers under square roots
- Consider $\sqrt{9} + \sqrt{4} = 3 + 2 = 5$
 - This is not equal to $\sqrt{9+4} = \sqrt{13} = 3.60555...$



Examiner Tips and Tricks

If your calculator gives an answer as a surd, leave the value as a surd throughout the rest of your working.

This will ensure you do not lose accuracy throughout your working.

Simplifying Surds

How do I simplify surds?

- To **simplify a surd**, factorise the number using a square number, if possible
 - If multiple square numbers are a factor, use the largest
- Use the fact that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ and then work out any square roots of square numbers



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- E.g. $\sqrt{48} = \sqrt{16 \times 3} = \sqrt{16} \times \sqrt{3} = 4 \times \sqrt{3} = 4\sqrt{3}$

LOOK FOR FACTORS THAT ARE SQUARE NUMBERS

THE ANSWERS HERE $\sqrt{8}$ AND $\sqrt{720}$ IN SIMPLEST SURD FORM

$$\sqrt{8} = \sqrt{4 \times 2} = \sqrt{4} \times \sqrt{2} = 2\sqrt{2}$$

$$\sqrt{720} = \sqrt{144 \times 5} = \sqrt{144} \times \sqrt{5} = 12\sqrt{5}$$

- When simplifying multiple surds, simplify **each separately**
 - This may produce surds which can then be collected together
 - E.g. $\sqrt{32} + \sqrt{8}$ can be rewritten as $\sqrt{16}\sqrt{2} + \sqrt{4}\sqrt{2}$
 - This simplifies to $4\sqrt{2} + 2\sqrt{2}$
 - These surds can then be collected together
 - $6\sqrt{2}$
- You may have to expand double brackets containing surds
 - This can be done in the same way as multiplying out double brackets algebraically, and then simplifying
 - The property $(\sqrt{a})^2 = a$ can be used to simplify the expression, once expanded
 - E.g. $(\sqrt{6} - 2)(\sqrt{6} + 4)$ expands to $(\sqrt{6})^2 + 4\sqrt{6} - 2\sqrt{6} - 8$
 - This simplifies to $6 + 2\sqrt{6} - 8$ which gives $-2 + 2\sqrt{6}$



Worked Example

Write $\sqrt{54} - \sqrt{24}$ in the form $\sqrt{q}$ where q is a positive integer.



Your notes

Simplify both surds separately by finding the highest square number that is a factor of each of them

$$9 \text{ is a factor of } 54, \text{ so } \sqrt{54} = \sqrt{9 \times 6} = 3\sqrt{6}$$

$$4 \text{ is a factor of } 24, \text{ so } \sqrt{24} = \sqrt{4 \times 6} = 2\sqrt{6}$$

Simplify the whole expression by collecting the like terms

$$\sqrt{54} - \sqrt{24} = 3\sqrt{6} - 2\sqrt{6} = \sqrt{6}$$

$$\sqrt{6}$$



Your notes

Rationalising Denominators

Rationalising Denominators

What does rationalising the denominator mean?

- If a fraction has a denominator containing a surd then it has an **irrational** denominator

- E.g. $\frac{4}{\sqrt{5}}$ or $\sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{\sqrt{3}}$

- The fraction can be rewritten as an equivalent fraction, but with a **rational** denominator

- E.g. $\frac{4\sqrt{5}}{5}$ or $\frac{\sqrt{6}}{3}$

- The numerator may contain a surd, but the denominator is **rationalised**

How do I rationalise simple denominators?

- If the **denominator** is a **surd**:
 - Multiply **the top and bottom of the fraction** by the **surd on the denominator**

- $\frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}}$

- This is equivalent to multiplying by 1, so does not change the value of the fraction

- $\sqrt{b} \times \sqrt{b} = b$ so the denominator is no longer a surd

- Multiply the fractions as you would usually, and simplify if needed

- $\frac{a\sqrt{b}}{b}$

How do I rationalise harder denominators?

- If the **denominator** is an **expression containing a surd**:

- For example $\frac{2}{3 + \sqrt{5}}$



Your notes

- Multiply the **top and bottom of the fraction** by the **expression on the denominator**, but **with the sign changed**

$$\frac{2}{3 + \sqrt{5}} = \frac{2}{3 + \sqrt{5}} \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}}$$

- This is equivalent to multiplying by 1, so does not change the value of the fraction

- Multiply the fractions as you would usually (use brackets to help)

$$\frac{2}{3 + \sqrt{5}} = \frac{2(3 - \sqrt{5})}{(3 + \sqrt{5})(3 - \sqrt{5})}$$

- Expand any brackets, and simplify

$$\frac{2}{3 + \sqrt{5}} = \frac{6 - 2\sqrt{5}}{(3 \times 3) + 3\sqrt{5} - 3\sqrt{5} - (\sqrt{5}\sqrt{5})} = \frac{6 - 2\sqrt{5}}{9 - (5)}$$

- You can use the **difference of two squares** to expand the denominator quickly

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - (\sqrt{b})^2 = a^2 - b$$

- This is what makes the denominator **rational**

- Simplify

$$\frac{2}{3 + \sqrt{5}} = \frac{6 - 2\sqrt{5}}{4} = \frac{3 - \sqrt{5}}{2}$$



Examiner Tips and Tricks

If your answer still has a surd on the bottom, go back and check your working!



Worked Example



Your notes

Write $\frac{4}{\sqrt{6}-2}$ in the form $p + q\sqrt{r}$ where p , q and r are integers and r has no square factors.

Multiply the top and bottom of the fraction by the expression on the denominator, but with the sign changed

$$\frac{4}{\sqrt{6}-2} = \frac{4}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2}$$

Multiply the fractions as you would usually

$$\frac{4}{\sqrt{6}-2} = \frac{4(\sqrt{6}+2)}{(\sqrt{6}-2)(\sqrt{6}+2)}$$

Expand the brackets

The denominator can be expanded using the difference of two squares

$$\frac{4}{\sqrt{6}-2} = \frac{4\sqrt{6}+8}{(\sqrt{6})^2-(2)^2} = \frac{4\sqrt{6}+8}{6-4}$$

Simplify

$$\frac{4}{\sqrt{6}-2} = \frac{4\sqrt{6}+8}{2} = 2\sqrt{6}+4$$

Write in the form given in the question

$$4 + 2\sqrt{6}$$

$$p = 4$$

$$q = 2$$

$$r = 6$$